

DEEPAK KUMAR SINGH

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PROFESSIONAL SUMMARY

Computer Science graduate with hands-on experience in full-stack web development (React, Node.js, Tailwind CSS) and machine learning (TensorFlow, OpenCV, MediaPipe). Built and shipped real-time gesture-controlled visualization tools, an ML-based activity detection system, and a production e-commerce site.

EXPERIENCE

Software Developer Intern — Extion Infotech, Agra, India

June 2024 – August 2024

- Collaborated with the software development team on a machine learning predictive model built with Python and TensorFlow, aimed at enhancing medical imaging analysis.
- Worked on data preprocessing and pipeline improvements to support model training and evaluation.
- Partnered with engineers across the team to integrate the ML component into the broader application workflow, gaining exposure to real-world software development practices.

Web Development Training — Udemy (Remote)

- Completed self-paced coursework in the MERN stack (MongoDB, Express.js, React, Node.js), practicing both frontend and backend development through guided project work.

PROJECTS

3D Model Tracer

- Built an interactive system that lets users manipulate 3D models in real time using webcam-based hand-gesture recognition — supporting rotation, zoom, and movement via finger tracking.
- Implemented optimized gesture mapping and noise-reduction techniques for smooth, responsive motion control.
- Tech stack:** Python, MediaPipe, OpenGL, PyOpenGL

Human Activity Detection

- Developed a machine learning pipeline to automatically classify human activities (walking, running, sitting, standing) from sensor and video data.
- Tech stack:** Python, TensorFlow, Scikit-learn, OpenCV

E-Commerce Website — RustiqueMud

- Built a responsive e-commerce website for RustiqueMud Soap, a handcrafted soap brand, covering product listing, cart, and checkout flows.
- Tech stack:** React, Tailwind CSS, Node.js

SKILLS

Languages: C, Python, JavaScript

Frontend: React, Tailwind CSS, HTML, CSS

Backend & Databases: Node.js, MongoDB, MySQL

Machine Learning / Computer Vision: TensorFlow, Scikit-learn, OpenCV, MediaPipe, OpenGL

Core Competencies: Data Structures & Algorithms, OOP, Full-Stack Development, Responsive Design, Backend API Development, Model Training & Evaluation, SDLC, Debugging & Testing, Cross-functional Collaboration, Problem Solving

Languages Spoken: English, Hindi

EDUCATION

Bachelor of Engineering (B.E.) in Computer Science

Dr. Bhimrao Ambedkar University, Agra, India | September 2021 – September 2025

- CGPA: 7.52. Coursework: Data Structures, C, Python, DBMS, Artificial Intelligence & Machine Learning, Mathematics.

Intermediate (10+2)

Shandilya Public School, Varanasi, India | April 2018 – April 2020

- 72% aggregate, with a focus on Mathematics and Science.

High School

Shandilya Public School, Varanasi, India | April 2016 – April 2018

- 84% aggregate, with a focus on Mathematics and Science.

CERTIFICATIONS

- Generative AI — LinkedIn Learning
- Web Analytics — Udemy
- Web Designing — Mindluster
- C Programming Competition — C-PROGRAMIIZ (2022)